Florida International University

Spring 2026 Senior Design Project

F!TZ

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PROBLEM

Many individuals struggle with their sense of style and often do not know how to create outfits using the clothing they already own. Despite having a full wardrobe, it can be difficult to visualize different combinations or experiment with new styles without purchasing additional items. As a result, users tend to repeat the same outfits or feel like they have “nothing to wear.” Additionally, there is a lack of accessible tools that allow users to organize their clothing and efficiently explore outfit options based on their existing wardrobe.

CAPSTONE I PROGRESS

During Capstone I, our team established foundational infrastructure across six major system components: mobile app frontend, image processing pipeline, clothing understanding (ML), closet database, outfit generation engine, and feedback loop. The image processing component implemented background removal using rembg with multi-format support (JPEG, PNG, HEIC) for cross-platform compatibility. A modular Python backend architecture was configured to support independent development of image processing, ML integration, and outfit generation logic. Regular team stand-ups coordinated integration points between frontend, backend APIs, and ML pipeline. Key technical decisions included virtual environment isolation for dependencies, FastAPI for RESTful communication, and a preprocessing workflow outputting standardized images for ML classification. These efforts position our team for full-system integration and user testing in Capstone II.

CAPSTONE II GOALS

The primary objective for Capstone II is to achieve full system integration and deploy a functional digital closet application. Key goals include completing the image processing pipeline with cropping and normalization modules, training and deploying the ML clothing classification model, implementing the outfit generation algorithm with compatibility scoring logic, and developing a seamless mobile interface for photo capture and outfit browsing. The team will establish backend infrastructure with user authentication, database integration via Supabase, and RESTful API endpoints connecting all components. Integration and user testing will validate system performance and recommendation quality, with iterative refinements based on feedback. The final deliverable will demonstrate end-to-end functionality from clothing photo upload to AI-powered outfit suggestions at the senior showcase.

SOLUTION

F!TZ is a virtual closet application that helps users organize their wardrobe and make better use of the clothing they already own. The platform allows users to view all their items in one place, explore different outfit combinations, and experiment with new styles without needing to buy more clothes. By simplifying how clothing is organized and presented, F!TZ makes it easier for users to get dressed, feel confident in their choices, and discover new ways to wear what they already have.

OBJECT DESIGN

For Capstone II

IMPLEMENTATION

For Capstone II

VERIFICATION

For Capstone II

SUMMARY

For Capstone II

REFERENCES

For Capstone II

ACKNOWLEDGEMENT

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